

Splunk Enterprise System Monitoring

Installation, configuration, SPL analysis and dashboard evidence

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05.08.2026



SPLUNK

Raw machine data becomes usable security insight

The project demonstrates an end-to-end monitoring workflow rather than a single tool feature.

1

Deploy

Install Splunk Enterprise and confirm Splunk Web is accessible.

3

Investigate

Use SPL to filter, aggregate and interpret relevant activity.

2

Ingest

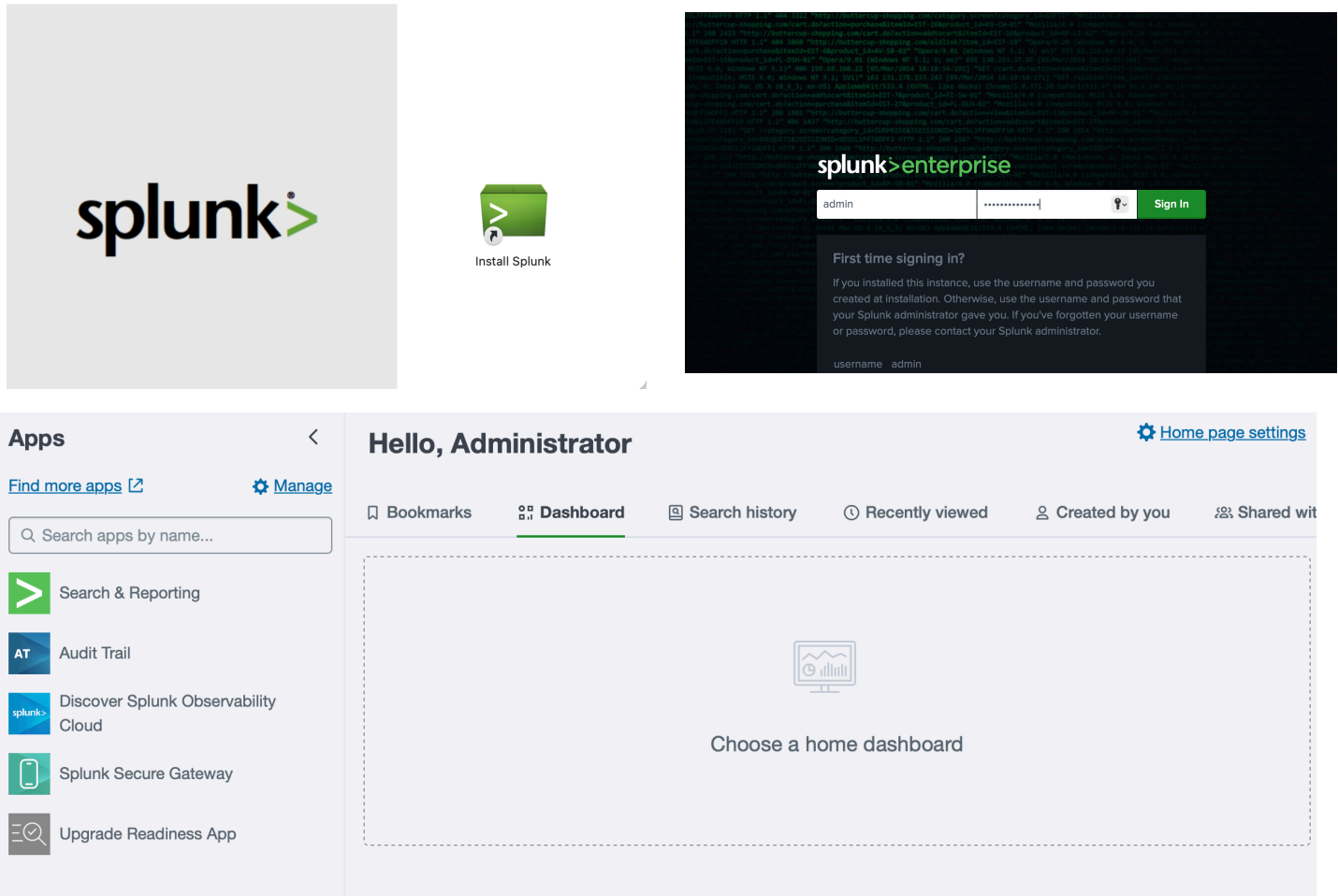
Configure log inputs so machine data is collected and searchable.

4

Communicate

Build a dashboard and summarise findings, limitations and lessons.

Installation established a reliable local analysis environment



Installation evidence

- Splunk Enterprise 10.4.2 installation package opened on macOS
- Splunk Web login page loaded successfully
- Administrator authentication completed successfully
- Search & Reporting application available for analysis

Figure 1. Installation, authentication and successful access to the Splunk Enterprise environment.

A small toolchain supported analysis and reporting

01

Splunk Enterprise
10.4.2

Search, indexing and
dashboards

02

Safari

Access to Splunk Web

03

macOS

Local Host System

04

Unix/Linux Add-
on

Version 10.3.2
Inputs: /Library/Logs and /
var/log

Local macOS system logs were collected through enabled file and directory inputs

Configuration made macOS system logs searchable

- 1

Install the add-on

Installed Splunk Add-on for Unix and Linux 10.3.2
- 2

Enable log inputs

Enabled /Library/Logs and /var/log in the app configuration

Install - Success

Splunk Enterprise installed **Splunk Add-on for Unix and Linux** (10.3.2).

Read Release Notes

View Apps

Done

File and Directory Inputs:

Name	Enable (All)	Disable (All)
/etc	☐	☒
/home/*.bash_history	☐	☒
/Library/Logs	☒	☐
/root/.bash_history	☐	☒
/var/adm	☐	☒
/var/log	☒	☐

Figure 2. Successful add-on installation and enabled macOS system-log inputs

Configuration made macOS system logs searchable

3

Verify event collection

Ran index=* and confirmed that indexed system events were returned

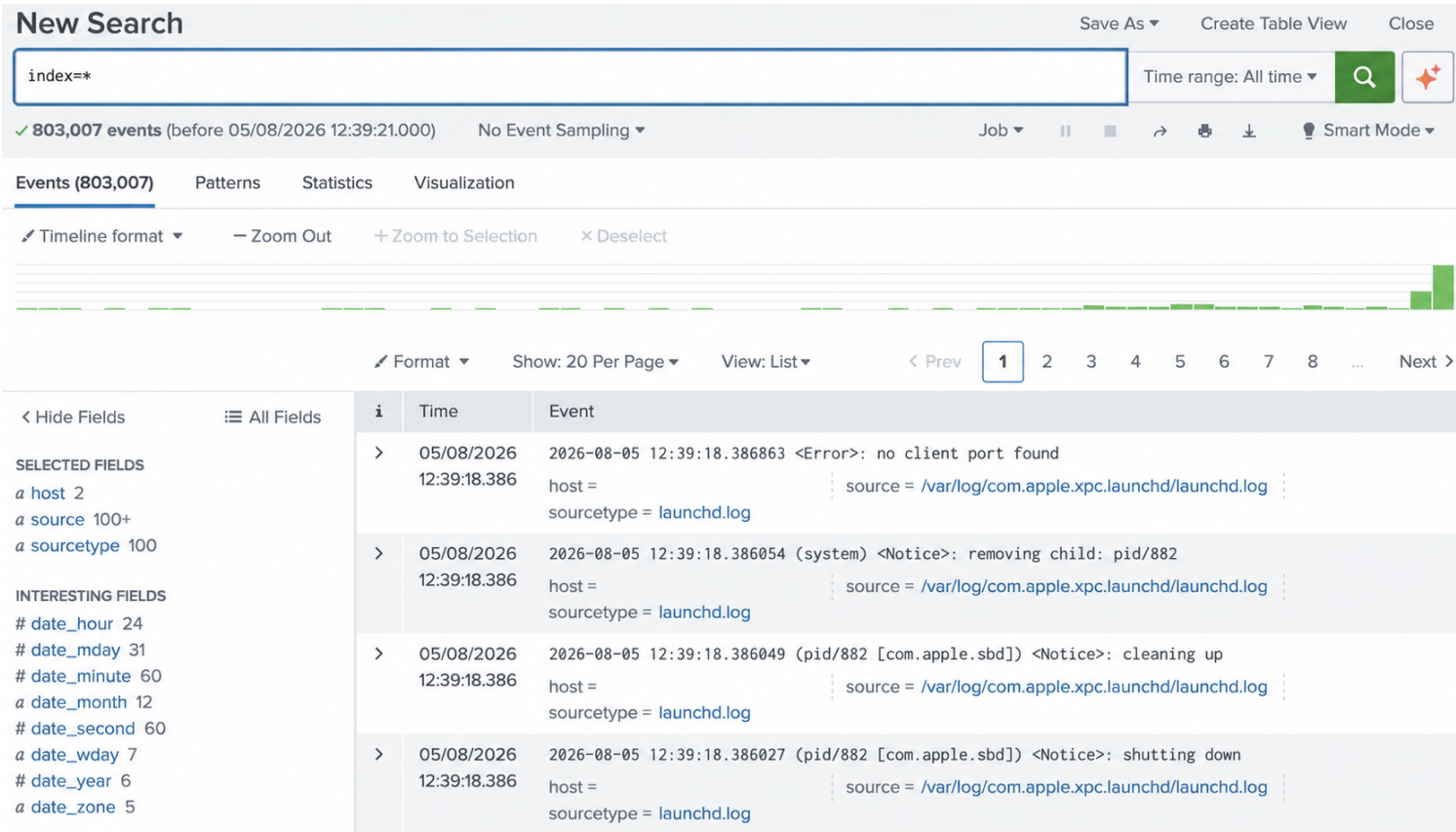


Figure 3. Validation search showing indexed events from the configured macOS log inputs.

SPL progressed from validation to focused analysis

1 Identified the highest-volume log source types

```
index=* | stats count by sourcetype
```

Description: This search grouped all indexed events by source type and counted events in each category. It showed which source types generated the highest event volumes during the selected 24-hour period.

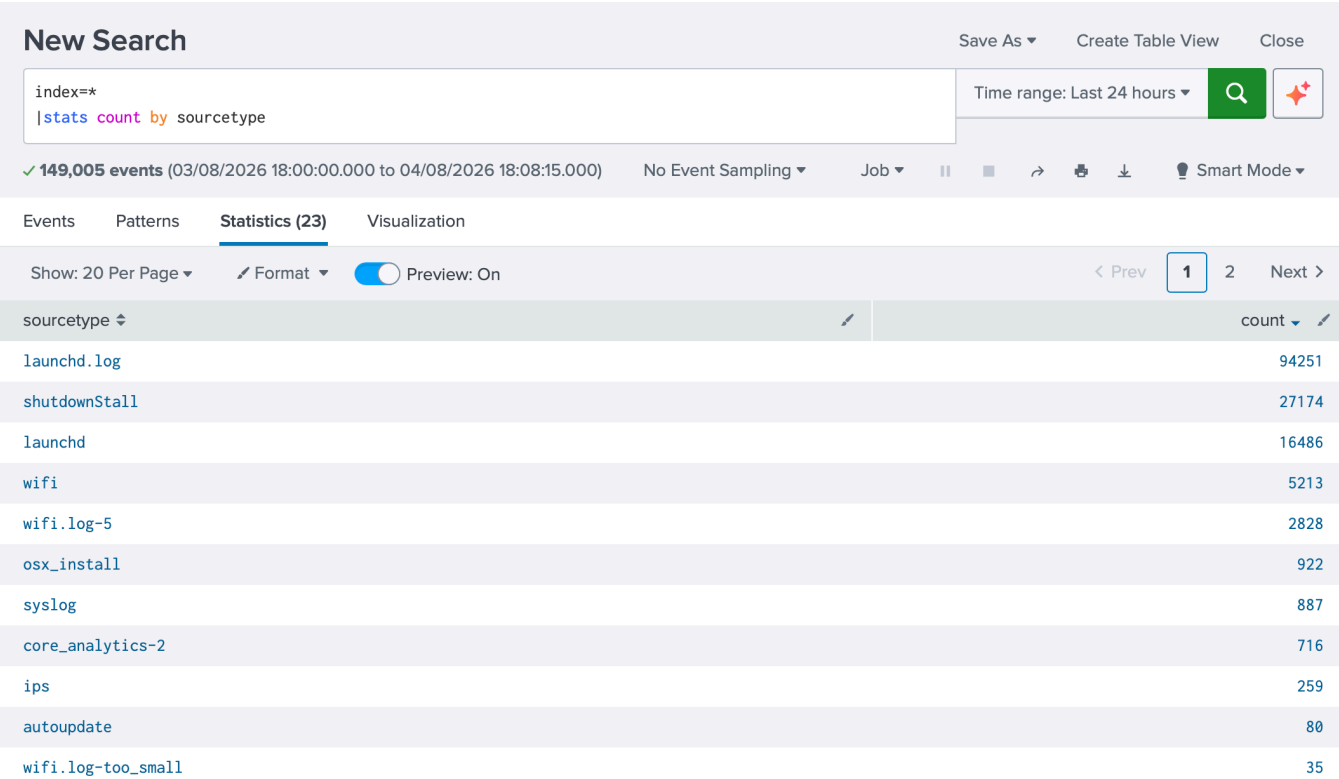


Figure 4. SPL results showing event counts grouped by source type.

SPL progressed from validation to focused analysis

2 Creating the visualisation

```
index=* | stats count by sourcetype | sort - count | head 10
```

Description: The original search returned too many source types for a clear visualisation. So I filtered it to display only the largest 10 categories. Producing a more focused chart that made differences in event volume easier to compare.

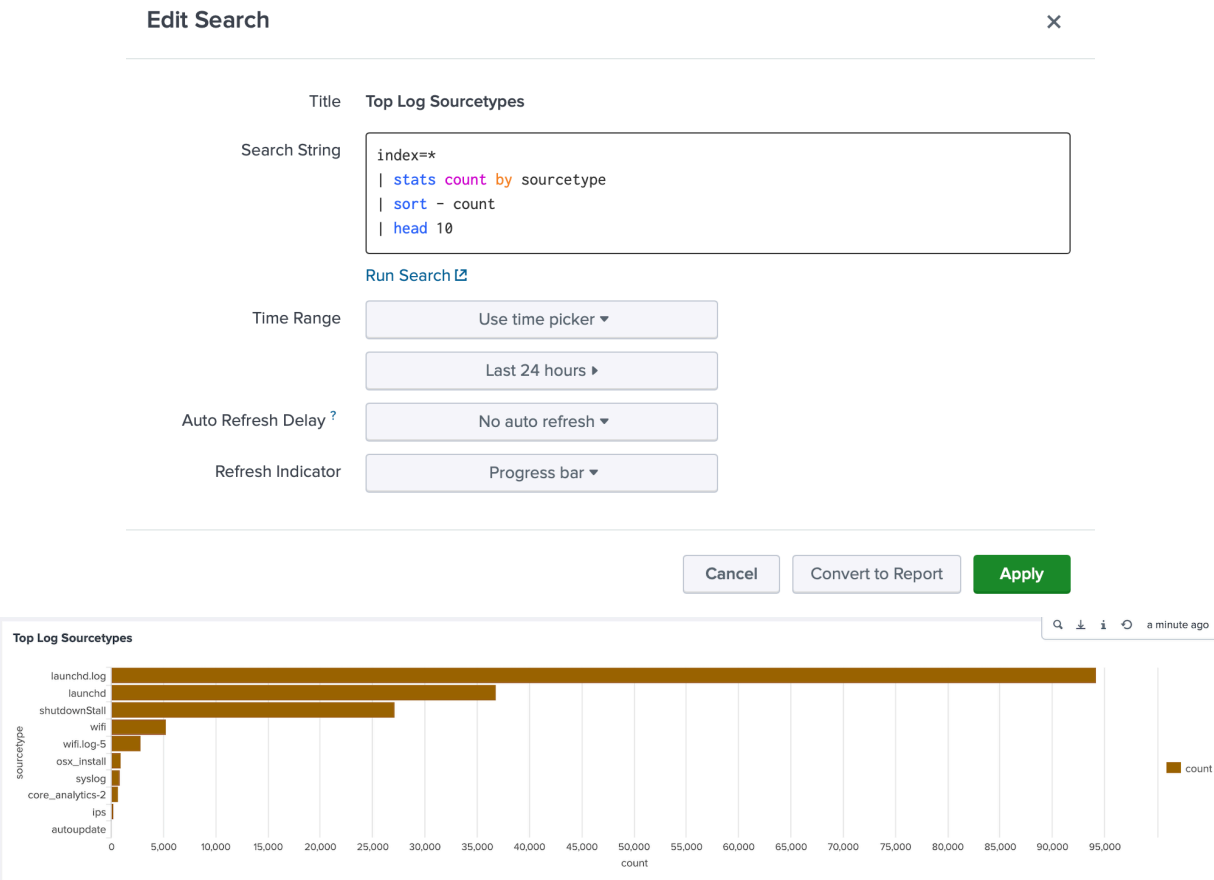


Figure 5. Refined SPL search and resulting top-ten source-types bar chart.

SPL progressed from validation to focused analysis

3 Creating two additional visualisations

Identifying the highest-volume Sources

```
index=* | stats count by source | sort - count | head 10
```

Description: This search counted events by their individual log source and ranked the ten highest volume sources. It helped confirm which configured files were actively supplying data and resulted in the bar-chart seen on the right.

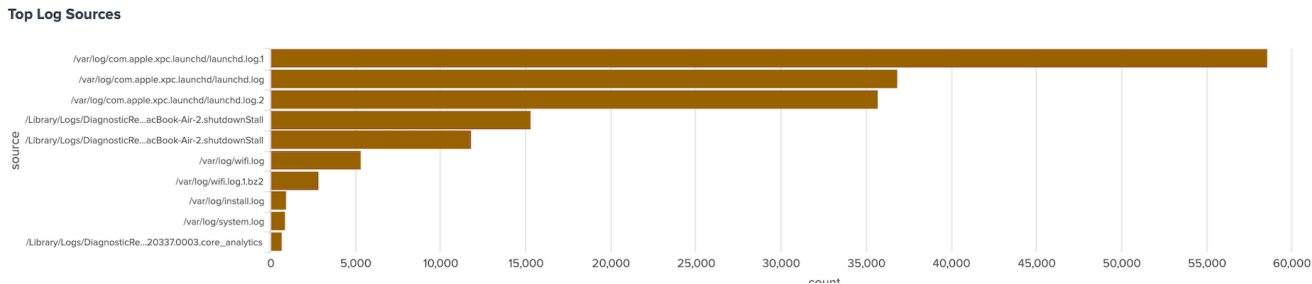


Figure 6. Refined SPL search and resulting Top Log bar chart.

Potential issues by source Type

```
index=* (error OR failed OR warning) | stats count by sourcetype
```

Description: This search filtered indexed events containing the terms error, failed or warning and counted them by source type. Highlighting which log categories produced the most potentially significant events and resulted in the pie chart on the left

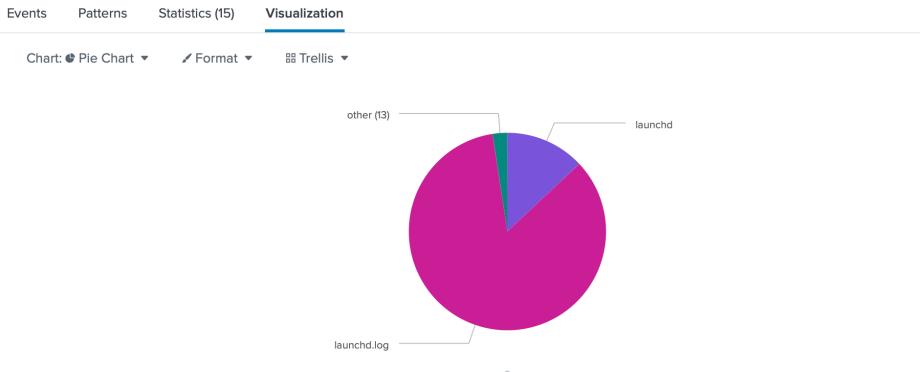


Figure 7. Potential issue events grouped by source type.

The dashboard combined three searches into one monitoring view

Description: The dashboard centralised event distribution, source-type volume and individual log-source activity in one view.

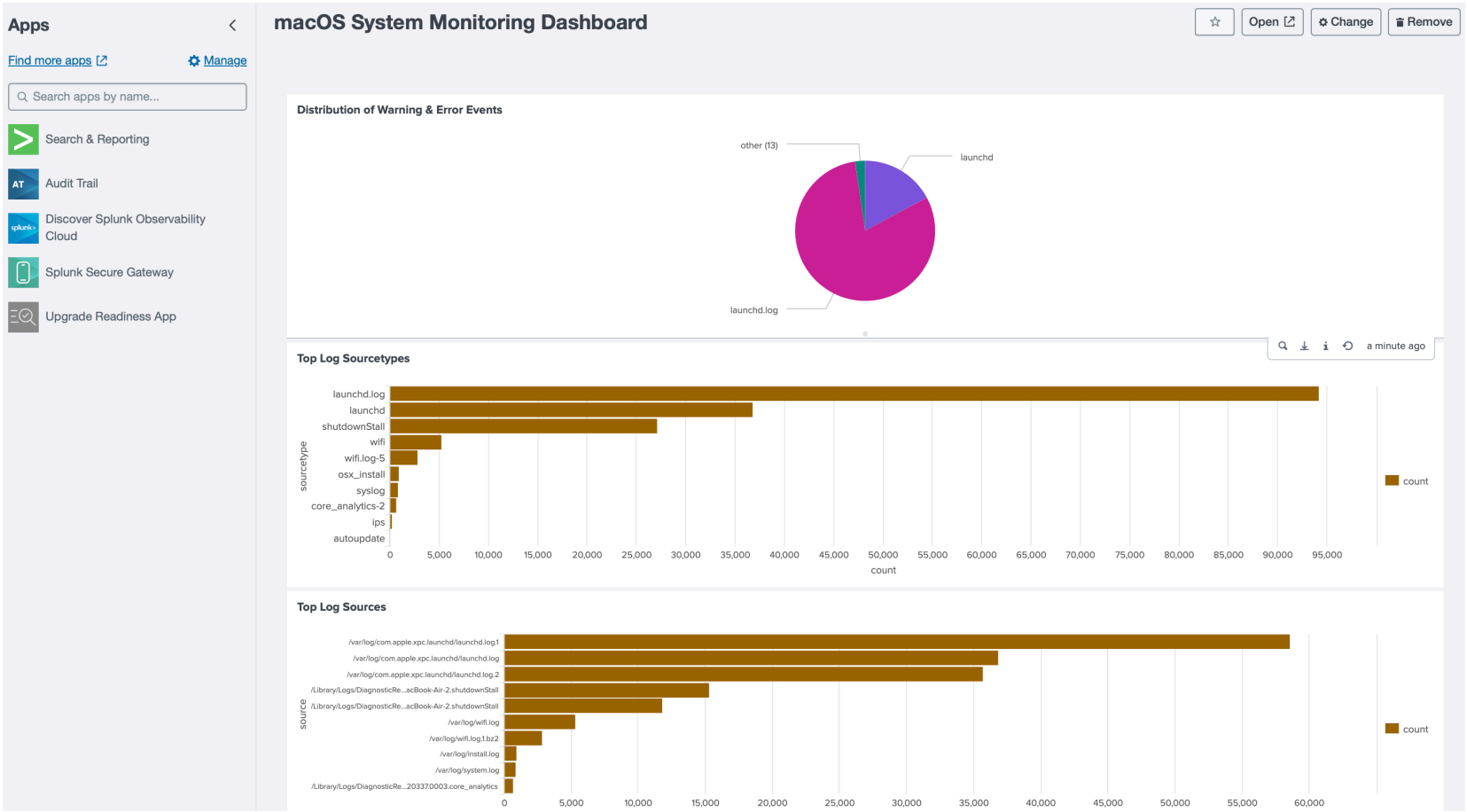


Figure 8. Completed dashboard combining event volume, top log sources and potential issues

Frequent shutdown-stall events warranted further investigation

- | | | |
|----|------------------------|---|
| 01 | Most important pattern | ShutdownStall was the second-highest-volume source type, indicating repeated events associated with delayed or incomplete shutdown activity. |
| 02 | Supporting evidence | During the selected 24-hour period, ShutdownStall produced 27,174 events-more than the 16,486 events recorded for the general launchd source type. |
| 03 | Security relevance | Repeated shutdown stalls could affect system availability and may indicate misconfiguration, unresponsive services or processes that are not terminating correctly. The related processes, timestamps and nearby system events should be reviewed to determine the cause. |

Launchd activity dominated the monitored system logs

- | | | |
|----|------------------------|---|
| 01 | Most important pattern | Launchd-related logs generated substantially more events than the other monitored source types. |
| 02 | Supporting evidence | Over the selected 24-hour period, launchd.log produced 94,251 events and also dominated results containing error, failed or warning terms. |
| 03 | Security relevance | High event volume does not by itself indicate an attack. However, repeated warnings or errors could indicate service instability, misconfiguration or unusual process activity, so the event messages, processes and timestamps should be investigated further. |

Data quality and iteration shaped the strongest lessons

What worked

- Validate ingestion with index=* before beginning the detailed analysis
- Refining broad search results with sort and head to create clearer visualisations
- Saving three useful searches as panels within one monitoring dashboard

What I would improve

- Create a dedicated index instead of searching across every available index
- Define required log sources, source types and time ranges earlier
- Add alert thresholds and inspect the underlying events rather than relying only on error-related keywords

The project established a repeatable monitoring workflow

Splunk Enterprise was installed and configured to collect, search and visualise macOS system log data.

- Installation, authentication and data ingestion were successfully verified
- SPL searches identified high-volume launchd activity and repeated shutdown-stall events
- Three visualisations were combined into a reusable monitoring dashboard
- Further development could introduce a dedicated index, improved event filtering and automated alerts